Experimental Results and Analysis

1 Experimental Results & Analysis

1.1 Accuracy Analysis Across Perception and Reasoning

1.1.1 Perception and Accuracy Reasoning

Table 1 reveals a clear distinction between perception and reasoning capabilities across the evaluated MLLMs. Mistral Small 3.2 achieves the highest APS accuracy in most settings, indicating a strong ability to perceive, interpret, and convert multimodal mathematical information into structured triple representations. But, the best APS accuracy across models and noise variants (76.23%) still leaves substantial room for improving perceptual capability. Moreover, high perception accuracy does not always translate into superior final-answer accuracy: Llama 4 Scout often performs better on reasoning, suggesting stronger downstream reasoning mechanisms. This highlights that effective perception alone is insufficient for solving mathematical problems and must be complemented by robust reasoning.

Across modality settings, all models exhibit sensitivity to input representation, although the observed trends are not uniform. Under clean conditions ($I + T$ or FTFD), perception accuracy is generally higher when text and diagrams are provided separately (Setting-2). Under noisy conditions, Gemma 3 4B and Llama 4 Scout typically show improved perception performance in Setting-2 across both lossless and lossy perturbations, whereas Mistral Small 3.2 exhibits a mostly declining or inconsistent trend. In contrast, reasoning accuracy decreases for Gemma and Llama in Setting-2, while Mistral Small 3.2 demonstrates improved reasoning performance. These contrasting behaviors suggest that different MLLMs process and reason over multimodal information in fundamentally different ways. Additionally, across noise variants, perception accuracy is generally higher under lossless perturbations than under lossy ones, as expected. Surprisingly, this pattern does not consistently hold for reasoning accuracy, where lossy variants sometimes outperform lossless ones. A plausible explanation is that lossless perturbations may introduce redundant triples, whereas lossy variants can yield more concise representations that are easier to reason over.

1.1.2 Accuracy Retention

The Accuracy Retention (AR) scores in Table-1 suggest that Llama 4 Scout is the most consistent model, retaining the highest proportion of correctly answered clean-condition questions under noisy settings, followed by Mistral Small 3.2, while Gemma 3 4B exhibits the lowest accuracy retention and the greatest sensitivity to noise.

1.1.3 Cross Model Perception and Reasoning Performance Evaluation

To reduce bias from model-specific reasoning ability, we perform cross-model accuracy evaluation by using triples generated by one MLLM as input to other MLLMs. Specifically, we conduct this cross-model analysis between the best and worst-performing models (Llama 4 Scout and Gemma 3 4B, respectively). This helps isolate perception quality from reasoning quality, providing a fairer assessment of the generated triples. From the cross-model heatmap fig. 1, Llama 4 Scout appears to outperform Gemma 3 4B on both the perception and reasoning components. When Llama-generated triples are used as input to Gemma's reasoning module, accuracy is consistently higher than when Gemma 3 4B reasons over its own triples. Conversely, when either model's triples are provided to Llama 4 Scout for reasoning, Llama 4 Scout achieves substantially better accuracy than Gemma 3 4B. These results empirically demonstrate that Llama 4 Scout has stronger perceptual extraction and reasoning capabilities, suggesting that larger models may possess more robust perception, richer structural representations, and enhanced mathematical reasoning ability.

1.1.4 Ablation Study

In the APS ablation study, we find that for triples generated under Setting-1, directly using the source question often yields higher perception accuracy than using the derived triples, whereas the results in Setting-2 are more mixed. This indicates that input modality substantially influences perception. A plausible explanation is that vision–language models are primarily trained on natural images and captions, and are therefore more effective at processing raw visual or natural-language inputs than structured triple representations.

Comparing perception and reasoning across models, Gemma 3 4B and Mistral Small 3.2 exhibit relatively strong perception but weaker downstream reasoning, while Llama 4 Scout performs well on both. Moreover, perception and reasoning performance vary notably across models, suggesting that model architecture and scale play an important role in shaping perception and reasoning capabilities, consistent with the trends observed in Table 1 and the cross-model heatmap.

1.2 Tracability Analysis

To evaluate traceability, we selected 50 APS questions (including both correctly and incorrectly answered cases) and traced back the perception triples to localize error sources.

For correctly answered APS questions, the model typically identified the key perception triples needed to solve the question. It occasionally retrieved a few irrelevant triples, but the MLLM effectively identified them as noise. For instance, when determining whether points E, O, and D form a triangle, the model produced the correct answer despite additionally retrieving an irrelevant triple (F, `lies_on`, `circle`).

During error traceability for incorrectly answered APS questions, we found that the model frequently selected inappropriate triples to answer a question, even when suitable triples were present in the candidate set. In addition, the model often generated triples that were either redundant or semantically incorrect, leading to incorrect APS answers. For example, in an APS item asking which point lies on segment AD, the Mistral Small 3.2 retrieved the triples (`ADC`, `connects`, D), (`ADC`, `connects`, C), and (`ADC`, `connects`, A), and then incorrectly selected point D instead of C. These triples are redundant because $\angle ADC$ corresponds to a straight line, and the model should instead rely on simpler, more informative relations such as (C, `connects`, A) or (C, `connects`, D). Using depth-first search (DFS), we traced the error to node D in the triple (`ADC`, `connects`, D), indicating that both triple generation and triple selection contributed to the failure.

In another APS question asking for the length of a line segment, Mistral Small 3.2 selected the triples `(AB, has_measure, 3.2 units)`, `(AC, has_measure, 8.0 units)`, `(BC, parallel_to, CD)`, and `(AB, perpendicular_to, BC)`. Backtracking revealed that the value `8.0 units` associated with AC was incorrect, and this erroneous measurement propagated to the final APS answer. Additionally, semantic matching by the reference MLLM correctly aligned semantically equivalent relations such as `has_measure` and `is_length`.

1.3 Summary of Analysis and RQ Answers

Taken together, the APS accuracy results in Table 1, the cross-model heatmap (Fig. 1), and the traceability analysis in Section 1.2 suggest that structured semantic triples provide a clear, but only partially reliable, representation of multimodal perception, thereby answering **RQ1**. The observed APS accuracies indicate that MLLMs possess meaningful perception capabilities and can successfully structure a substantial portion of the information present in text and images. However, final reasoning accuracy remains relatively low across all settings. Since the extracted triples contain the information required to answer the questions, this gap suggests that the challenge lies not only in perception but also in reasoning over structured representations. One possible explanation is that MLLMs are predominantly trained on natural language and visual inputs, making subject–predicate–object triples a less familiar representation. Consequently, models may successfully extract information yet struggle to effectively utilize it for downstream reasoning. To overcome this bottleneck, future frameworks may need to align structured representations more closely with natural language or explicitly train models to reason over knowledge graph triples.

The analysis in Subsection 1.1.1 further answers **RQ2**, demonstrating that modality representation significantly influences how different MLLMs perceive and utilize multimodal information. Moreover, the effects of modality representation are further shaped by the type and severity of noise introduced within each modality.

Finally, the joint analysis of APS accuracy, reasoning accuracy, and cross-model performance answers **RQ3**. While higher-quality triple representations generally lead to better downstream performance, strong perception alone does not guarantee successful reasoning. The superior cross-model performance of Llama 4 Scout demonstrates that downstream accuracy depends not only on perception quality but also on a model's ability to effectively reason over the extracted knowledge. These findings indicate that perception and reasoning are related but distinct capabilities, both of which are necessary for successful multimodal mathematical problem solving.

Table 1: Final-answer accuracy (majority-vote) and APS perception accuracy under different modality settings

Sub-setting	Model	Final Answer (Acc., AR)						APS (Acc.)					
		$I+T$	$I+T'$	$I'+T$	I_h+T_h	$I_h+T'_h$	I'_h+T_h	$I+T$	$I+T'$	$I'+T$	I_h+T_h	$I_h+T'_h$	I'_h+T_h
Setting-1	Gemma 3 4B	7.41	5.37, 51.25	6.15, 58	4.07, 35	5, 42.50	4.52, 42	36.34	34.84	34.04	30.59	26.24	27.55
	Mistral 3.2 24B	15.56	13.52, 56.55	13.56, 56.67	15.56, 54.76	16.67, 54.76	17.26, 57.62	74.45	76.33	76.23	65.34	65.19	65.32
	Llama 4 Scout	22.59	22.31, 70.08	22.96, 72.79	21.11, 60.66	21.49, 68.03	23.26, 71.48	54.20	54.88	63.36	60.12	42.86	47.37
Setting-2	Gemma 3 4B	6.67	4.44, 26.39	5.926, 28.89	2.96, 22.22	3.33, 12.50	3.18, 23.33	55.70	55.73	55.28	44.39	38.86	42.10
	Mistral 3.2 24B	17.41	15.65, 61.7	15.04, 60.85	17.78, 57.45	15.28, 53.72	15.52, 50.64	78.78	75.71	74.82	60.39	61.63	62.62
	Llama 4 Scout	24.44	20.92, 66.29	21.70, 66.67	17.41, 51.52	19.91, 57.20	20.37, 56.67	70.03	56.45	66.22	53.05	48.34	51.78

$I+T$: clean image and clean text (FTFD); $I+T'$: clean image with lossless text; $I'+T$: lossless image with clean text; I_h+T_h: lossy image with lossy text; $I_h+T'_h$: lossy image with losslessly perturbed lossy text; I'_h+T_h: losslessly perturbed lossy image with lossy text.

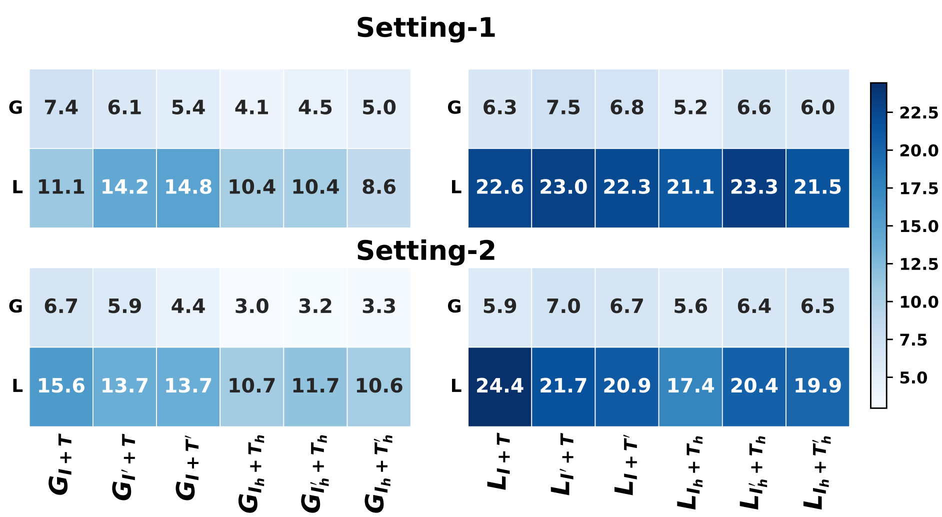

Figure 1: Cross-model evaluation of perception triples. Columns denote the triple generator model and settings; rows denote the reasoning model. G and L denote Gemma 3 4B and Llama 4 Scout, respectively; e.g., G_{I+T} denotes triples generated by Gemma under the $I+T$ or FTFD input, while G/L on the rows indicate the reasoning model.

Table 2: APS accuracy vs final-answer accuracy using the original question and generated triples.

Setting	Model	Final-Answer Accuracy		APS Accuracy	
		Triples	Original Question	Triples	Original Question
Setting-1	Gemma 3 4B	5.31	21.11	36.34	80.0
	Mistral Small 3.2 24B	15.3	25.56	74.75	91.13
	Llama 4 Scout	22.52	68.89	54.20	88.63
Setting-2	Gemma 3 4B	4.31	25.56	55.70	56.2
	Mistral Small 3.2 24B	15.33	26.29	78.78	74.3
	Llama 4 Scout	20.78	51.48	70.03	65.0